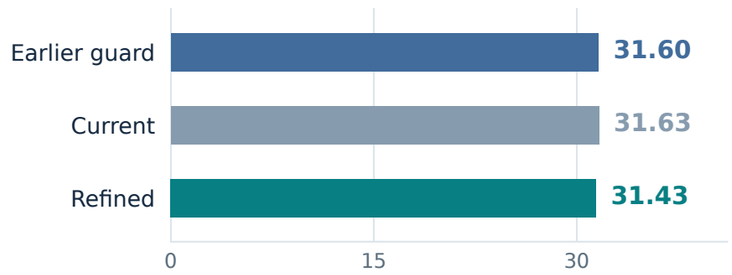
Narrow signed coverage repairs two retained tradeoffs

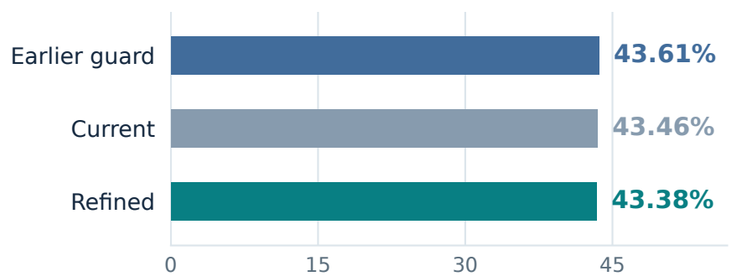
32 new training circuits · one fixed 300-tree fit · signed data/Boolean AST routes · four stored forests in the composite policy

A Consumed research panel · 192

MAE · LUT4 · lower is better

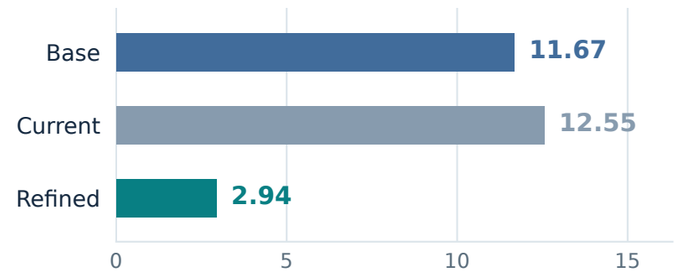


Positive-target MAPE · lower is better

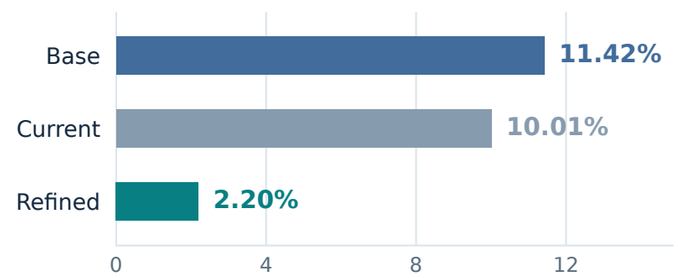


B Consumed signed add/Boolean · 4

MAE · LUT4 · lower is better

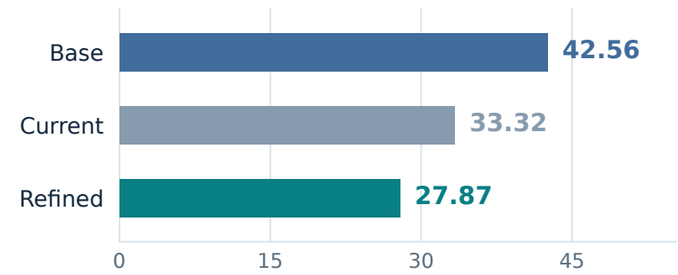


Positive-target MAPE · lower is better

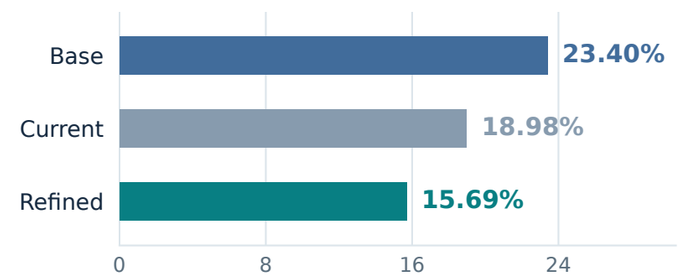


C Independent fresh check · 40

MAE · LUT4 · lower is better



Positive-target MAPE · lower is better



All seven retained panels: MAE, MAPE and worst underprediction no worse than current; prior fixes retained.

Frozen gates passed.

Fresh budget mistakes are unchanged: base = current = refined

Cap · LUT4	False accept	False reject	Predicted / actual feasible
32	**0**	**0**	**0 / 0**
64	**1**	**0**	**4 / 3**
128	**9**	**0**	**22 / 13**

Worst underprediction · LUT4

120.50 → 116.08 → 115.00

Base → current → refined

Points stay ≥ base; decisions at 32 / 64 / 128

retain 1 false accept at 64 and 9 at 128.

New subgroup regression: compound signed add/XOR (4 cases), current → refined MAE 26.005 → 26.441; MAPE 15.680% → 15.776%.

Inherited controls, earlier guard → current = refined: offset/sub MAE 66.49 → 76.05, MAPE 31.19% → 34.70%; static-signed mixed MAE 6.49 → 8.31, MAPE 7.56% → 9.77%.

Primary selected on consumed development data before fresh labels; fallback reported separately. Four stored forests and structural routing form the policy.

Earlier 192 / 327 / 32 / 40 / 40 / 32 / 40 panels are consumed. Fresh40 is separate; no learning curve or universal subgroup improvement is implied.

Experimental isolated iCE40 LUT4 points only. No calibrated safety, CPU, timing, power or application-transfer guarantee. Base stays the default.